



Connected Healthcare, BAN and Weaponised Pacemakers



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6 3,895,000,000. That is how many rupees are expected to be spent by healthcare providers in India on devices, telecom services, data centres and information technology (IT) in 2015 alone, according to Gartner.

Medical electronics is a lot more exciting now, with electronic components that resorb after use as well as stents that sense blood flow and temperature that these then transmit for analysis and storage. While you can read about more such amazing innovations in the article on page 26, this article gives you a glance at the electronics behind the scenes—components, devices, standards and other elements that power these innovations to life.

Imaging and diagnostic applications

In the east, China's Nanjing University is working on an acoustic diode technology that could improve the performance of ultrasound systems. Similar to the usual diodes that allow current only in one direction, this diode

allows sound in just one direction, thus eliminating acoustic disturbances caused by sound waves going in both directions.

Another recent addition for those into designing ultrasound systems is the availability of production processes and tools for full-scale capacitive micromachined ultrasonic transducers (cMUTs). These allow ultrasound imaging to enjoy higher lateral and axial resolution, higher speed and even real-time 3D imaging for diagnostic applications for areas such as ophthalmology and dermatology.

On the other side of the planet, University of Michigan's Wireless Integrated MicroSensing and Systems (WIMS²) Centre has made progress in its work on magnetoelastic sensors. By applying the devices into the circulatory system and bile ducts it is able to detect biliary sludge on a stent in the bile duct or tissue accumulation on arterial stents in cardiology.

EBR Systems has developed an implantable externally-powered grain-sized